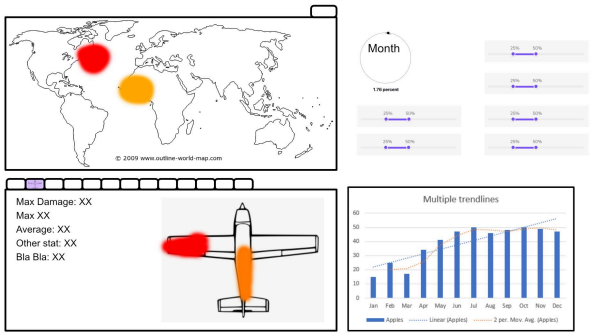
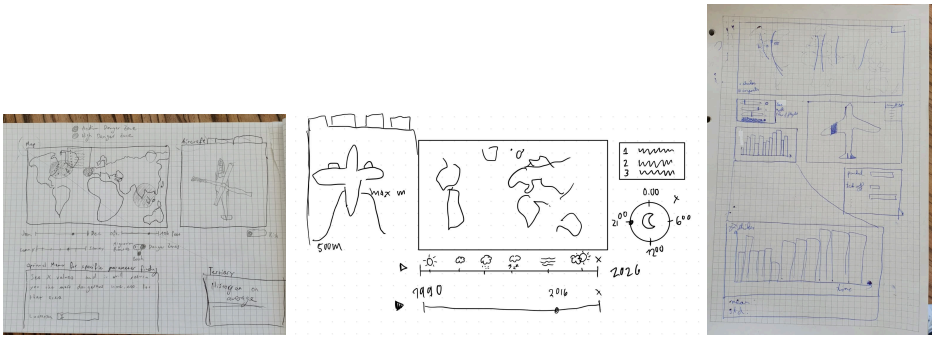
M2 - Project Goal - ThreeLittleBirds

Functional Prototype: https://three-little-birds-nine.vercel.app/

Sketches

Each team member first made their own sketch of their idea of how everything should come together and then we aggregated our ideas in a final sketch.

Final Sketch	
Individual Sketches	

Tools

Our main visualization is going to be an interactive map that displays the locations of the different bird strikes. The main tools that will be needed are D3.js for the interactive aspects of the visualization which allow us to dynamically display the data based on the sliders, and to toggle on and off the main migration routes. There will be points that mark the locations of the bird strikes, and colored areas indicating the areas with more strikes are. Furthermore, we plan to use libraries such as OpenLayers to create and visualize the background. We will need the concepts learned in the interactions lecture and the upcoming maps lecture to implement these features, as well as more general concepts learned throughout the course to aid in designing the final visualization.

The secondary visualization will be closely tied to the main visualization and is all about the damage done to the aircraft. The data displayed in this visual will change based on the interaction of the user with the main visualization. It will be a window with a few different tabs that can be switched from and to. The first tab will aggregate basic information that is

displayed in the main visualization. The other tabs will show a “blueprint” view of each aircraft type in the dataset that in certain areas will be coloured in a heatmap way, based on the location and severity of the damage, the aircraft type received during the bird strikes. These heatmaps will be created using SVG images where we will map areas to the dataset. How to create SVG was introduced in the Web Dev lecture. For this secondary visual the lectures on JavaScript and D3 events, when hovering over the heatmap of an aircraft, are of course important and the lectures about perception of colour and mark channels as well, so that the heatmap can be easily understood and interpreted.

The tertiary visualization is going to be a dynamic bar chart with aggregation. Its purpose is to deliver basic statistical information such as mean and median. The idea is to have a fixed y-axis which is the number of strikes and a choosable x-axis. It then displays the data based on the x-axis but also takes into account the other factors that are selected through sliders used for the main visualization. This visualization complements the main map visualization. D3.js provides all the tools needed to implement this visualization. Additionally the lectures on data and interactions are going to be very helpful.

Finally, we will rely on the designing visualisation lecture as a base for the evaluation of our visualisations and we will need to take into account the lecture about storytelling, when the visualizations will come together, so that the story of our page is well structured.

Pieces to implement

The main visualization can be broken down into a few steps that need to be achieved. First, the geographic data needs to be properly displayed and lined up with the background map. Secondly, we need to add the interactive elements that allow for a dynamic representation of the data. Then, we need to compute and display the “danger zones”. And finally, we are looking to incorporate a toggle that will display the bird migration paths from the secondary dataset. A bonus, minor feature that could be included if time permits, is the addition of a label that can indicate information from the strike when hovering over the points on the map. Also a “Play” button could be implemented that auto-scrolls through the years.

The absolute minimum for the secondary visual is that the general damage data is aggregated in the first tab and each aircraft type has its own damage heatmap on the whole dataset. Still, this is not enough and the interaction between the main visualization and the secondary visualization must be implemented as well, meaning that only the datapoints of the “Strike Hotspot Map” that are currently displayed are actually used for the heatmaps in the individual damage profiles. A helpful feature to implement would be that when hovering over heatmap sections, more detailed data about the specific section is displayed in a small infobox. An additional helpful feature would be that more than one aircraft type could be displayed at the same time, making comparisons between the different types easier. Also a toggle to a “Financial Damage” view could be interesting, that would display a heatmap based on the cost of strikes.

The minimal viable product of the tertiary visualization consists of a bar chart implemented using D3.js. In this initial version, the y-axis represents the number of strikes, while the x-axis is fixed to a single variable e.g. time. The chart is updated dynamically to changes with the sliders. Several enhancements can further improve this visualization. The first of

them being to make the x-axis choosable, allowing users to explore different dimensions of the data. Additionally, incorporating mean and median into the plot to provide deeper insights. Finally, animations can be added for smooth transitions when the plot gets updated through manipulation of the sliders or changes in the x-axis. However, these animations are primarily aesthetic and therefore have lower priority compared to the analytical features.